

HK DSE Biology — Photosynthesis (detailed test notes)

1. Overview

Photosynthesis converts light energy into chemical energy in green plants, algae, and some bacteria. Carbon dioxide and water become glucose and oxygen. Word equation: carbon dioxide + water $\rightarrow$ glucose + oxygen (light, chlorophyll).

2. Where it happens

Leaf palisade cells are packed with chloroplasts. Chlorophyll a and b in the thylakoid membranes absorb red and blue light and reflect green.

Stomata on the lower epidermis let CO₂ in and O₂ and water vapour out.

Xylem brings water from roots. Phloem exports sugars as sucrose.

3. Light-dependent stage (thylakoid / grana)

Photolysis splits water: $2\text{H}_2\text{O} \rightarrow 4\text{H}^+ + 4\text{e}^- + \text{O}_2$. Electrons enter photosystem II, then photosystem I. NADP⁺ is reduced to NADPH. ATP is made by chemiosmosis through ATP synthase (photophosphorylation). Oxygen diffuses out via stomata.

4. Light-independent stage (Calvin cycle, stroma)

Rubisco fixes CO₂ onto ribulose biphosphate (RuBP). The product is reduced using ATP and NADPH to triose phosphate. Some triose becomes glucose; the rest regenerates RuBP. Temperature, CO₂ concentration, and light intensity can each limit the rate (limiting factors).

5. Limiting factors (exam points)

- Low light: fewer electrons and less ATP/NADPH, so Calvin cycle slows.
- Low CO₂: rubisco has less substrate; extra light cannot raise the rate.
- Low temperature: enzyme (rubisco) collisions slow; rate falls.

A graph of rate vs light rises then plateaus when CO₂ or temperature limits.

6. Adaptations of the leaf

Broad lamina captures light. Thin blade shortens the diffusion path.

Waxy cuticle cuts water loss. Air spaces in the spongy mesophyll hold CO₂.

Guard cells open stomata in light when photosynthesis uses CO₂.